Ken Tanaka Hernández

PhD Researcher | HPC & AI Engineer | Scientific Computing

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"We are a way for the cosmos to know itself." — Carl Sagan, Cosmos: A Personal Journey (1980)

Professional Summary

Researcher specializing in turning complex research into real-world products. Background across **physics, mathematics, and computer science**, building scalable platforms with **HPC, Big Data, and AI**. Architecting **distributed systems** and **parallel workflows** for high-throughput research and enterprise workloads. Translating academic research into production systems, roadmaps, and stakeholder value with expertise in **physics-informed AI** for trustworthy, explainable, and robust predictions.

Experience

Istituto Nazionale di Oceanografia e di Geofisica Sperimentale (OGS) & Università degli Studi di Trieste (UniTS) — Trieste, Italy

PhD Researcher — Nov 2023 – Present

- Developed automated seismic data processing pipelines on **HPC clusters** using **SLURM, MPI, and GPU parallelization**.
- Implemented **ML-based earthquake detection** workflows integrating **SeisBench, PyOcto, and NonLinLoc**.
- Created dynamic job submission systems that allocated workloads across compute nodes with **32+ cores/node**.
- Designed reproducible scientific workflows using **GNU Make and Hydra** configuration frameworks.
- Achieved **94% detection accuracy** on production seismic workloads across Italy's northeastern national seismic network.

ORACLE de México, SA de CV — Guadalajara, Mexico

Software Developer II — Jul 2020 – Aug 2022

- Architected and deployed a **production backend** for automated analytics through a customer-facing API.
- Engineered automated patch recommendation workflows to streamline customer-facing diagnostic analysis.
- Mentored junior developers and established **code review practices**.

Hewlett Packard (HP) Labs — Guadalajara, Mexico

Intern, Modeling & Numerical Simulation — Jun 2019 – Jun 2020

- Developed thermal-process simulations for Multi Jet Fusion 3D printing technology.
- Validated computational models against experimental materials-science benchmarks.

Universidad de Guadalajara (UdeG) — Guadalajara, Mexico

Research Assistant — Apr 2018 – Jun 2020

- Conducted **computational physics** research on complex systems and statistical mechanics.
- Developed **Monte Carlo simulations** for phase transition analysis in social dynamics models.

Education

Università degli Studi di Trieste (UniTS) — Trieste, Italy

PhD in Applied Data Science & Artificial Intelligence (ADSAI) — Nov 2023 – Present

Built **distributed training/inference pipelines** across **multi-node GPU clusters (MPI/CUDA)** for large seismic datasets; engineered a custom validation framework using **bipartite optimization** and weighted similarity scoring.

Expected graduation: Oct 2026. Completed concurrently with MHPC studies.

Scuola Internazionale Superiore di Studi Avanzati (SISSA) — Trieste, Italy

Master in High Performance Computing (MHPC) — Sep 2023 – Aug 2024

Developed an **end-to-end ML pipeline** for real-time earthquake detection; architected **HPC-optimized workflows** with **Transformers and CNNs** to ingest continuous data streams at scale.

Graduated with honors. Completed concurrently with PhD studies.

International Centre for Theoretical Physics (ICTP) Trieste, Italy

Postgraduate Diploma in Quantitative Life Science (QLS) Sep 2022 – Aug 2023
Studied **Approximate Message Passing (AMP)** for rank-one estimation in the spiked Wigner model; developed and evaluated a **prior-independent scheduler** for the Onsager correction term. Applied the method to unstructured and structured random-matrix noise, reproducing Bayes-optimal behavior in the standard model and identifying limitations for structured noise.

Universidad de Guadalajara (UdeG) Guadalajara, Mexico

BSc. Physics Aug 2016 – Dec 2021
Simulated complex social systems with **1D Ising spin models** integrating Kawasaki and Schelling dynamics; quantified critical exponents via **finite-size scaling** ($z \approx 2.48$, $\lambda \approx 1.15$, $\theta \approx 0.46$).
Graduated with postgraduate studies. Completed concurrently with BEng studies.

Instituto Tecnológico y de Estudios Superiores de Occidente (ITESO) Guadalajara, Mexico

BEng. Nanotechnology Aug 2016 – Dec 2018
Coursework concurrently with BSc studies.

Technical Skills

Core	**HPC, Big Data & Distributed Systems**, Mathematical Modeling, **Physics-Informed AI**, Scientific ML, Data Engineering, Backend Architecture, MLOps, Parallel Algorithms, API Design
Programming	**Python** (NumPy, SciPy, Pandas, PyTorch, TensorFlow, CuPy, Dask), **C/C++**, SQL (PostgreSQL, Oracle), JavaScript/TypeScript, Advanced Bash (sed/regex, GNU Make)
Deep Learning	**Transformers, CNNs, RNNs, GANs, PINNs**
HPC	**SLURM, MPI/MPICH, OpenMP, OpenACC, CUDA**. Dynamic thread/task optimization (32+ cores/node). CINECA (Galileo100, Leonardo), ICTP Ulysses
Seismology	**SeisBench, NonLinLoc, PyOcto, REAL**. Waveform processing, phase picking, station metadata, seismic catalogs
Infrastructure	**Linux, Docker, Git, CI/CD**, environment management, job monitoring, error/output logging
Methods	Pipeline orchestration, **Hydra YAML, GNU Make**, Agile/Scrum, TDD, Technical Documentation
Languages	Spanish (native), English (C1), Italian (A2), Japanese (conversational), Chinese (A1)

Selected Projects

Automated Seismic Data Processing Pipeline

End-to-end HPC pipeline for seismic event detection: waveform acquisition, ML-based phase picking (SeisBench), octree-based association (PyOcto), and 3D hypocenter location (NonLinLoc). Dynamic SLURM job generation with template-based configuration and GNU Make orchestration.

Event Matching & Validation Framework

Novel validation system using **Hungarian algorithm-based bipartite optimization** with custom weighted similarity metrics (location, magnitude, origin time). Enables rigorous comparison against authoritative catalogs (INGV). Open-source contribution for reproducible seismology research.

Additive Manufacturing Thermal Simulator

High-fidelity **finite element simulation** of powder bed fusion processes. Validated against experimental thermal imaging data from HP's Multi Jet Fusion production line.

Social Dynamics Simulation Engine

Monte Carlo framework for studying segregation and opinion dynamics in complex networks. Implemented Ising-based models with Kawasaki-Schelling hybrid dynamics. Characterized critical exponents via finite-size scaling.

Publications

Insights into a Deep Learning Workflow for Automatic Detection of Seismic Signals at Mount Etna Volcano

Carducci, A.; Cocina, O.; Sciotto, M.; Cannata, A.; Di Gioia, S.; Vuan, A.; Saraò, A.; **Tanaka Hernández, K.**; Sugan, M. *Seismological Research Letters*, 2026. DOI: 10.1785/0220250431.

Unpublished / Forthcoming

Evaluating an AI-enabled HPC workflow for seismic monitoring in the Southeastern Alps

Tanaka Hernández, K.; Münchmeyer, J.; Magrin, A.; Di Gioia, S.; Vuan, A.; Picozzi, M.; Rodriguez Garcia, A.; . . . ; Sugan, M. *Manuscript in preparation.*

Comparative Evaluation of Clustering Algorithms for Seismic Event Association: Assessing their suitability for reliable event association

Tanaka Hernández, K., Rodriguez Garcia, A., Vuan, A. *Manuscript in preparation.*

Conferences

Performance assessment of Deep Learning picking models at Mount Etna volcano

Carducci, A., Cocina, O., Sciotto, M., Cannata, A., Di Gioia, S., Vuan, A., Saraò, A., **Tanaka Hernández, K.**, and Sugan, M.: *EGU General Assembly 2026*, Vienna, Austria, 3–8 May 2026, EGU26-11585, DOI: 10.5194/egusphere-egu26-11585.

Benchmark tests of Deep Learning pickers at Mount Etna volcano

Carducci, A.; Cocina, O.; Sciotto, M.; Cannata, A.; Di Gioia, S.; Vuan, A.; Saraò, A.; **Tanaka Hernández, K.**; Sugan, M. (2026). *GNGTS – Gruppo Nazionale di Geofisica della Terra Solida, 44th National Conference of the GNGTS*, 10–13 February 2026, Udine, Italy.

Artificial Intelligence (AI) and Machine Learning (ML) on Big Data seismology

Tanaka Hernández, K.; Münchmeyer, J.; Magrin, A.; Di Gioia, S.; Rodriguez Garcia, A.; Sugan, M. (2026). *GNGTS – Gruppo Nazionale di Geofisica della Terra Solida, 44th National Conference of the GNGTS*, 10–13 February 2026, Udine, Italy.

Artificial Intelligence (AI) and Machine Learning (ML) on Big Data Seismology

Tanaka Hernández, K. (2025). *24th Mexican International Conference on Artificial Intelligence (MICAI 2025) Doctoral Consortium*, 3–7 November 2025, Guanajuato, Mexico.

Datasets

Revised OGS Seismic Catalog for Northern Italy (2005–2024) Using 1D Velocity Model with NonLinLoc: Quality-controlled, homogeneous dataset for seismological and machine learning applications

Magrin, A.; **Tanaka Hernández, K.**; Sugan, M. *Zenodo*, forthcoming dataset.

Honors & Awards

2023–2024	**TeRABIT Scholarship** — **HPC** training and Earth-science research	OGS
2022–2023	**Postgraduate Diploma Fellowship** — Competitive **Theoretical Physics** scholarship	ICTP
2020	**3rd Place, National Poster Presentation Contest** — Research poster competition	HP

Volunteering

ISC High Performance — Science and Technology

Student Volunteer — Jun 2025

Engaged with leading minds in high performance computing, including renowned scientists, early-career researchers, and representatives of organizations advancing HPC.

Université Grenoble Alpes (UGA) — Science and Technology

Research Internship — Apr 2025 – Jul 2025

Universidad de Guadalajara (UdeG), CUCEI — Education

Science Outreach: — Jan 2017 – Dec 2019

- **Noche de las Estrellas (2017, 2018, 2019)**
- **Circo de la Ciencia**
- **Others**

Presented experiments and explained scientific phenomena to encourage community engagement with science; answered questions from observers.

Hewlett-Packard (HP) Education

Hour of Code Jan 2020
Partnered with local schools to teach coding basics to young girls and underrepresented groups, helping inspire the next generation of reinventors.

Agencia Espacial Mexicana (AEM) Science and Technology

Student Volunteer Sep 2016
Assistant at the 67th International Astronautical Congress (IAC) 2016.

Hobbies & Interests

- **Resilience & Well-being:** Mindfulness, swimming, cycling, running, and gym training.
- **Creativity & Innovation:** Photography and research in biology, physics, mathematics, and computer science.
- **Learning & Exploration:** Traveling, Chinese, philosophy, financial literacy and personal investments.
- **Community Engagement:** Science outreach, volunteering, and mentoring.

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